The petrology, geochemistry, and remanent magnetization of metabasalts and metagabbros in the Lac Patu – Fort McKenzie region, New Quebec Territory

MAURICE K.-SEGUIN
Département de Géologie, Université Laval, Québec (P.Q.), Canada G1K 7P4

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One hundred and five specimens were collected from 40 different sites in northwest-striking metabasalts and metagabbros of Lower Proterozoic (Aphebian) age in the northeastern part of the Labrador Trough (longitude: 69° W, latitude: 57° N). The radiogenic age determinations of surrounding rocks of the same age yielded 1860 m.y. A petrological and chemical description of the rocks is included. The chemical results for total Fe, MgO, and $Na_2O + K_2O$ combined, drawn on a ternary diagram show that the metabasalts and metagabbros fall in the tholeiite field. Magnetite (with some occasional 5% ulvospinel) is the most frequent magnetic memory carrier; pyrrhotite is less commonly found. In order to obtain some pertinent information relative to the stability of the NRM component, step-wise alternating field demagnetization was conducted on 40% of the specimens and the others were demagnetized at a specific AF intensity. In 35% of the specimens, the AF demagnetization process induced an anhysteretic magnetization component above 250 Oe. A secondary isothermal remanent magnetization of lesser stability was often detected in the 50–150 Oe range. Tilting of the formations sometimes increased and sometimes decreased the dispersion of the NRM vectors. After tilting the lithological units that were AF demagnetized, the mean paleomagnetic pole position is 152° E, 20° S (dm = 14°, dp = 07°, K = 34), with at least one reversal.

Cent cinq specimens orientés ont été obtenus de 40 localités différentes dans des métabasaltes et des métagabbros d'âge Protérozoïque Inférieur (Aphébien) dans le secteur centre-nord de la fosse du Labrador (longitude: 69° W, latitude: 57° N). L'âge relatif des roches avoisinantes déterminé par méthode K–Ar est de 1860 Ma. On a inclus une étude pétrologique et géochimique; les analyses chimiques démontrent que les métabasaltes et les métagabbros appartiennent au clan des tholéites. Le minéral opaque le plus commun est la magnétite, mais la pyrrhotine est aussi présente. De manière à obtenir des renseignements relatifs à la stabilité de l'aimantation rémanente naturelle, on a procédé à un lavage par champ alternatif progressif sur 40% des échantillons. Au cours de ce lavage, on a observé la présence d'une composante anhystérétique au-delà de 250 Oe dans 35% des cas. On a aussi détecté la présence d'une composante isothermale secondaire dans l'étude de 50–150 Oe. Le degré de dispersion a tantôt augmenté tantôt diminué lors du dépliage des formations. Après dépliage et nettoyage par champ alternatif, la position moyenne des paléopoles établis pour les localités est 152 °E, 20 ° S (dm= 14 °, dp= 07 °, K= 34) et au moins un reversement des pôles est évident. Il est plus que probable que cette position du paléolopole obtenue en faisant la moyenne des pôles aux diverses localités n'est guère significative. La dispersion des pôles d'une localité à une autre suggère que la rémanence a originé à diverses périodes de temps aux différentes localités.

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Introduction

This study is focussed on the paleomagnetism and geochemistry of the metabasalts and metagabbros of the northeastern portion of the Labrador geosyncline. More precisely, the area investigated is located in the Lac Patu (longitude: 69°00′–69°15′ W; latitude: 56°30′–56°45′ N) and Fort McKenzie areas (longitude: 68°30′–69°00′ W; latitude: 56°45′–57°00′ N). The objective of this paper is therefore twofold: firstly to obtain chemical, petrographic, and paleomagnetic data of the above-mentioned rock types; secondly, to derive their pole position from stability-tested data. Since paleomagnetism is becoming of greater importance in determining the time sequence of Precambrian rocks and in studying the nature of Precambrian orogenies, many additional paleomagnetic studies ought to be conducted in order to determine the variations of the Precambrian geomagnetic field. This study is a small contribution to this general requirement, but it is of some importance, since no definite trend of apparent polar wandering has yet been firmly established for the 1200–2000 m.y. span (McGlynn *et al.* 1974; McGlynn and Irving 1975). Such uncertainties require new

TABLE 1. Average chemical analyses of different rock types investigated in this study, expressed in wt%

	Fine grained metabasalts	Fine grained metagabbros	Medium grained metagabbros	Coarse grained metagabbros
No. of samples	42	28	39	27
SiO_2	46.93	48.92	48.55	48.49
TiO_2	1.98	1.40	1.42	1.08
Al_2O_3	12.30	13.64	13.95	14.55
Total Fe	12.23	12.59	11.54	10.95
MgO	9.12	6.63	6.86	6.75
MnO	0.17	0.21	0.17	0.20
CaO	8.01	9.65	9.19	9.61
Na_2O	2.54	1.98	2.75	2.86
K_2O	0.50	0.49	0.46	0.72
P_2O_5	0.21	0.10	0.12	0.08
H_2O^+	3.12	2.32	3.07	2.62
H_2O^-	0.51	0.13	0.10	0.13
CO_2	0.52	0.18	0.34	0.16
Fe^{++}	8.53	9.59	8.30	8.63
Cu	0.01	0.01	0.01	0.02
Ni	0.05	0.02	0.01	0.01
S	0.25	—	0.12	0.05
Total	98.45	98.28	98.67	98.30

paleomagnetic results from rock sequences magnetized during this period. In this respect, the Labrador Trough is an untapped source; a very limited amount of paleomagnetic data is available on the magnetic taconite and Montagnais diabase dykes of the Schefferville area (Seguin 1972*a*,*b*) and the magnetic itabirite of the Mount Wright – Carol Lake region (Seguin 1974*a*,*b*).

Geology

The region investigated lies within areas mapped by the Geological Survey of Canada at a scale of 1 in. = 4 mi (Roscoe 1957; Fahrig 1965) and by the Quebec Department of Natural Resources (Dressler 1973, 1974; Ott 1972). The study area lies entirely within the Labrador Trough, which is part of the Circum-Ungava geosyncline (Dimroth *et al.* 1970). The Aphebian Kaniapiskau Supergroup, which consists of sedimentary and volcanic rocks, rests unconformably on the Archean basement. The sedimentary rocks, mainly in the miogeosyncline (the western portion of the Trough), includes conglomerates, arkoses, sandstones, siltstones, argillites, cherts, cherty ironstones, and dolomites. The volcanic rocks of the eugeosyncline (the eastern portion of the Trough) include some basalts, gabbros, rhyolites, tuffs, and agglomerates, with intercalations of sedimentary rocks (argillites, siltstones, and minor conglomerates).

All the volcanic rocks of the eugeosynclinal sector were included in the Montagnais Group by Dressler (1974), even though part of this volcanic sequence may be classified in the Murdoch Formation (Baragar 1967). The Montagnais Group consists of metagabbros and metabasalts.

Petrology and Geochemistry

Table 1 shows the results of the chemical analysis of 16 principal components for four different rock types present in the region investigated. Forty-two samples of fine grained metabasalts, 28 fine grained metagabbros, 39 medium grained gabbros, and 25 coarse grained gabbros were chemically analyzed in this fashion.[1] Three samples belonging to the metabasalt, fine grained metagabbro, and coarse grained gabbro groups, respectively, yielded statistically differing values within their own group and are not representative of the lithologic unit.

The chemical analyses show that the chemical

[1]An appendix that lists all the individual chemical results used in this paper is available, at a nominal charge, from the Depository of Unpublished Data, CISTI, National Research Council of Canada, Ottawa, Ontario, Canada K1A 0S2.

composition of the metabasalts is significantly different from the one of the metagabbros. This difference is particularly apparent when comparing the slightly lower SiO_2—CaO and substantially higher TiO_2—MgO—H_2O—CO_2 content of the metabasalts with the one of the metagabbros. The $Na_2O + K_2O$ content amounts to 3% in both the metabasalts and metagabbros. The chemical composition of the fine, medium, and coarse grained metagabbros is not significantly different. Combining the results for the total Fe, MgO, and $Na_2O + K_2O$ content, it is found that the metabasalt and metagabbro samples fall in the tholeiite field.

The ferrous/ferric ratio is quite high for the metavolcanics and basic intrusives of the area, indicating that the igneous rocks are in a highly reduced state and that the oxidation state of the opaque minerals is low. The Fe^{++}:Fe ratio increases from 69 to 72, 75, and 79% for the metabasalts, fine, medium, and coarse grained gabbros, respectively. This information suggests that the metamorphism did not alter too drastically the primary chemical composition of these igneous rock types.

Age

A specimen of lamprophyre collected by B. Dressler in the Patu Lake area and termed meimechite has been sent for radiometric age determination to K. L. Currie of the Geological Survey of Canada. The result yields 1873 ± 53 m.y. (whole rock K–Ar). This age coincides very well with an age determination by Fryer (1972). His Rb–Sr whole rock isochron studies of slates of the Menihek Formation yielded an age of 1870 ± 50 m.y.

Sampling Procedure

In 1971, some 30 oriented specimens were collected at 16 different sites in the Lac Patu area and in 1973, some 75 additional oriented specimens were collected at 24 sites in the Fort McKenzie region. The number of samples per site varied between 1 and 4. In this regional survey, efforts were made to select sites where there was an excellent exposure and where the structural geology was rather simple and well understood. Great care was also taken to select fresh material. The sample locations were chosen by reference to detailed geological maps published by the Department of Natural Resources of the Province of Quebec (1 in. = 1 mi; Prelim. Rep. 603 and GM28318). As the formations are seldom anomalously magnetic, a magnetic compass could be used for the orientation of the rock samples. The samples were cut into small cubes having a side length of one inch.

Measurements

Magnetic susceptibility measurements were made with a standardized resistance bridge manufactured by Cenco Instruments Corporation, having a precision of 5×10^{-7} cgs emu. The direction and intensity of remanent magnetization was measured by means of a custom built, low revolution (5 Hz) spinner magnetometer as well as a high revolution Princeton Applied Research spinner magnetometer. Alternating field demagnetization was carried out to remove unwanted secondary components, using the demagnetizer built at Laval University and having a maximum peak-to-peak intensity of 1800 Oe. The Earth's magnetic field is cancelled with large Helmholtz coils and concentric mumetal cylinders, coaxial with the solenoid.

Magnetic Susceptibilities and Koenigsberger Ratios

Histograms of the volume ferromagnetic susceptibilities and Koenigsberger ratios were constructed. Figure 1 shows a histogram of the magnetic susceptibilities of the metabasalts and different types of metagabbros. The metabasalts and coarse grained metagabbros have a low magnetite content with respective modal values of 5×10^{-5} and 2×10^{-4} cgs emu/cm^3. The fine and medium grained metagabbros depict a bimodal distribution with a common modal value of 7.5×10^{-5} cgs emu/cm^3. The second modal value is 1.5×10^{-3} and 7×10^{-3} cgs emu/cm^3 for the medium and fine grained metagabbros, respectively. These relatively high susceptibilities are due to a mineralization of magnetic pyrrhotite, which is predominant in the fine grained gabbros. Figure 2 shows a histogram of the Koenigsberger ratio (Q) for the same lithologic units. The metabasalts and the fine grained metagabbros have, respectively, a low (0.2) and a relatively high (1.7) Koenigsberger ratio. The Koenigsberger ratio of the medium and coarse grained metagabbros is bimodal; the low mode is situated between 0.2

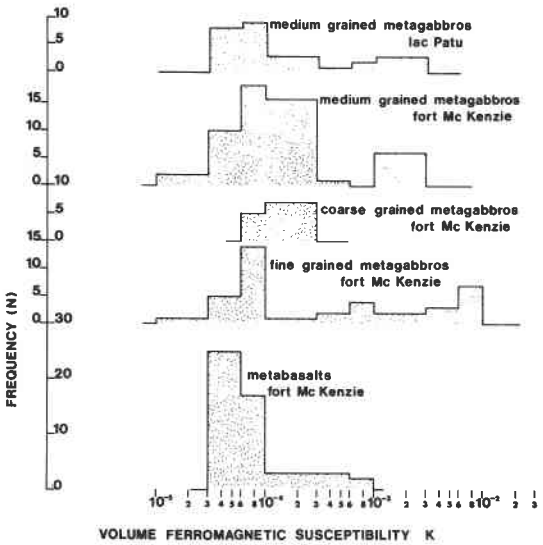


FIG. 1. Histogram of the volume ferromagnetic susceptibilities (measured in cgs emu/cm^3) for five different lithological units from the Lac Patu and Fort McKenzie areas.

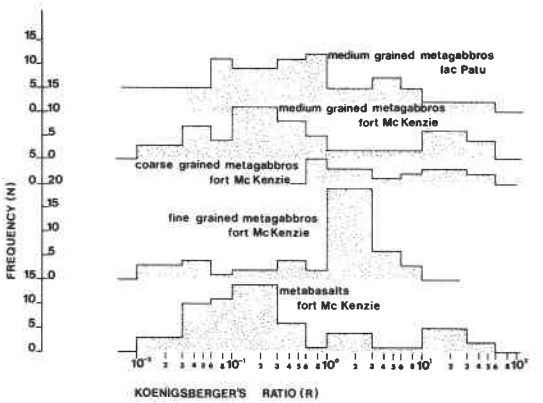


FIG. 2. Histogram of the Koenigsberger's ratios for the same lithological units as Fig. 1.

and 0.8 and the high mode between 5 and 20. Consequently, the NRM intensity is generally much larger in the metagabbros than in the metabasalts.

Alternating field demagnetization

At least one specimen from each of the 40 sites was demagnetized in steps of 50 Oe from 50 to 700 Oe and occasionally up to 1000 Oe in the absence of an ambient field. Guided by the step-wise changes in remanence (direction and intensity) of the pilot specimen(s) of the site, an intensity of AF demagnetization was selected using an objective 'end point' stability method, in which the remaining specimens from the site were AF demagnetized and their residual remanence measured (Symons and Stupavsky 1973). Alternating field demagnetization was performed on all specimens and the cleaned results were obtained in most instances after cleaning in fields of 100–250 Oe (Fig. 3). Normalized demagnetization curves for some typical test specimens from a few sites are shown on Fig. 4. Median destructive fields (mdf) are usually between 50 and 200 Oe. The magnetization directions of pilot specimens from 3 sites are presented in Fig. 5. All the other specimens were AF demagnetized at successive intensities of 100, 150, 200, and 250 Oe. The complete study involved approximately 1100 measurements of remanence.

Statistics and Grouping

The vector sum of the unit vectors representing the magnetization of all the oriented specimens from a given hand sample was calculated and taken as the mean direction of magnetization for that hand sample (Fisher 1953).

The grouping of the different sites was established on the basis of lithology and geographical location. On the basis of megascopic and microscopic descriptions, four lithologic units were distinguished (see Table 2); these are: (a) fine grained metabasalts (groups 4 and 8), (b) fine to medium grained metagabbros (groups

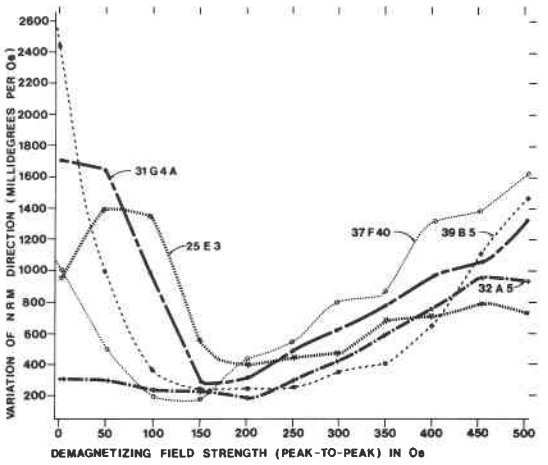


FIG. 3. Photo of the stability index for AF demagnetization of five typical specimens of metagabbros and metabasalts.

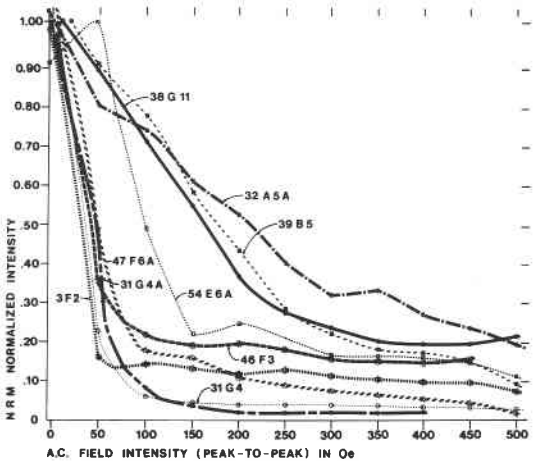


FIG. 4. Photo of normalized remanent intensity of nine typical specimens of metagabbros and metabasalts as a function of the demagnetizing field strength.

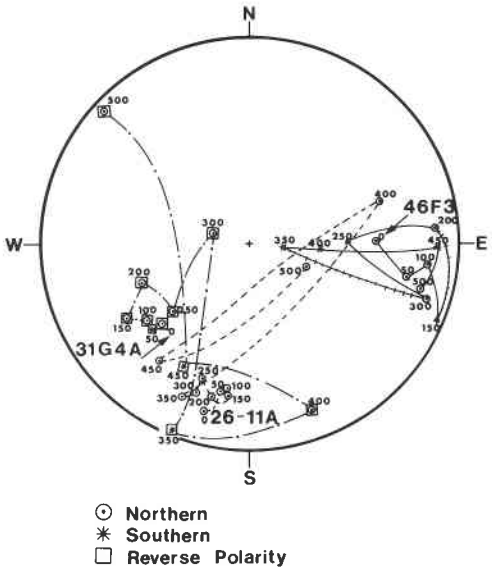


FIG. 5. Lambert Equal Area Plot, showing variation of the magnetization directions of three pilot specimens of metagabbros and metabasalts as a function of AF demagnetizing strength. Northern = North poles, Southern = South poles. Cross-hatched lines indicate change of hemisphere.

3 and 5), (c) medium grained metagabbros (groups 2 and 6), and (d) coarse grained metagabbros (groups 1 and 7). There are two geographical locations, which are the Lac Patu and Fort McKenzie areas, respectively.

Results of Paleomagnetic Measurements

The NRM directions of the different geological formations investigated showed variable dispersion. Before AF demagnetization, the mean direction of the NRM orientations of all geological formations is 096°, 19° ($\alpha_{0.95} = 18°$) before tilting of the formations and 109°, 01° ($\alpha_{0.95} = 24°$) after tilting. After AF demagnetization between 100 and 250 Oe (peak-to-peak), the dispersion within sites decreased slightly, but the dispersion between sites did not; the mean site directions became grouped about 054°, 22° ($\alpha_{0.95} = 29°$) before tilting of the formations. The residual component after AF cleaning is much smaller than the initial NRM intensity; the average ratio of the mean site intensities before and after cleaning at 150 Oe is 0.27. The mean direction of the AF demagnetized NRM orientations after unfolding is 119°, −11° ($\alpha_{0.95} = 30°$) with a corresponding mean paleopole position of 152° E, 20° S. The dispersion of the poles is still large after AF cleaning. On demagnetization, specimen magnetization directions were found to move slightly to the northwest, azimuths and sites that originally gave directions with low positive inclinations moved most commonly in such a way that stable end-points in direction yielded low negative inclinations. The stability of the remanence of the different formations investigated in this study is demonstrated by the departure of the site NRM directions from the recent Earth's field direction (332° E, 80° N) and by the evidence for reversal (*s*).

However, because many NRM vectors show very large within-site scatter and because there is also a large scatter when all the NRM directions of the sites of a lithological group are put together, the strongly scattered results ($\alpha_{0.95} \geq 30°$) of 35% of the hand samples constituting the 105 oriented specimens were removed. On this basis, all the hand samples of group 3 and almost all of group 5 were discarded (Table 2); this observation suggested that fine grained metagabbros are lithological units unsuitable for paleomagnetic studies. Similarly, very few hand samples of groups 2 and 6 (medium grained metagabbros) and group 7 (metabasalts) were acceptable for paleomagnetic studies. The scatter in NRM directions is smaller in coarse grained gabbros (group 8). It is also observed that there is no definite relation between an increasing number of specimens on a site and a decreasing angle of dispersion of the NRM directions, in-

TABLE 2. Statistical study of the paleopoles of the different rock types after AF demagnetization

Group no.	Site no.	Before tilting				After tilting		
		N	D_1(°)	I_1(°)	$\alpha_{0.95}$	D_2(°)	I_2(°)	$\alpha_{0.95}$
1	26BIF	4	018	63	16	172	47	09
	32A5A	3	090	−31	13	232	−71	07
Total and mean		7	053	41	32	184	17	37
Pole position: 174° E, 31° N			065° E, 24° S					
2	28A1B	3	293	12	57	317	12	48
	34A11A	2	277	−06	27	323	−42	26
	35A4A	2	—	—	—	—	—	—
	36A1A	3	—	—	—	—	—	—
	39B1	2	219	35	35	187	10	49
	39B5	2	088	33	21	041	50	36
	40B1	3	162	43	29	265	53	59
	40B2	3	116	25	34	140	72	57
Total and mean:		20	173	26	36	120	−26	33
Pole position: 75° E, 09° S			179° E, 17° S					
3*	30A3	2	270	−21	38	345	07	40
	30A13	3	012	−09	36	341	−42	43
Total and mean:		5	293	−23	32	299	−15	44
Pole position: 134° E, 02° S			149° E, 08° S					
4*	41B1B	3	041	19	34	064	82	41
	31A16	2	274	22	30	061	02	39
Total and mean:		5	068	−02	27	063	42	31
Pole position: 178° E, 58° N			167° E, 35° N					
5**	40F19A	2	—	—	—	012	26	50
	39E17A	3	024	51	39	014	52	48
	31G2	3	027	28	31	037	20	39
	33E22A	3	285	11	29	285	14	40
	38G11	3	043	71	23	218	49	33
Total and mean:		14	005	40	18	348	50	25
Pole position: 241° E, 43° N			268° E, 55° N					
6*	25E3	3	053	−30	39	048	53	38
	37F40A	3	312	30	40	351	72	35
	47F6A	4	294	01	09	294	02	10
	38G3A	3	021	70	21	029	42	29
	44F4A	4	295	−78	08	153	−68	07
Total and mean:		17	334	−01	27	344	26	31
Pole position: 276° E, 19° N			268° E, 34° N					
7*	54E6A	3	227	74	35	052	62	34
	54E4A	3	354	17	38	033	52	39
	3F2	4	092	37	40	294	36	48
	49E10A	2	006	05	07	353	04	06
	46F3	3	107	19	27	144	52	39
	50E12	3	235	20	23	332	75	24
Total and mean:		18	021	29	24	017	63	18
Pole position: 156° E, 33° N			212° E, 67° N					
8**	38G10	3	189	49	36	167	07	35
	31G4A	4	253	46	25	317	11	26
	46F4	3	308	−35	16	350	−17	17
	38F23	4	305	−61	04	289	26	03
	36E16	2	221	44	33	198	28	32
	28E11	4	141	−55	38	111	−46	42
	26-11A	4	188	11	30	188	−15	30
Total and mean:		24	229	−03	22	254	02	29
Pole position: 194° E, 22° N			172° E, 08° N					
Grand averages:		100	054	22	29	119	−11	30
Average pole position: 179° E, 23° N				152° E, 20° S				

*Mixed polarities, mainly reversed.
**Reversed polarities.

dicating an absence of stability of the NRM component rather than a statistical problem.

After removal of unreliable results corresponding to a large within-site scatter, the mean NRM orientation of the metabasalts (groups 4 and 8), the fine grained gabbros (group 5), the medium grained gabbros (groups 2 and 6), and the coarse grained gabbros (groups 1 and 7) are: 235°, 06°; 300°, 28°; 322°, 31°; and 282°, 28°, respectively. The unweighted mean NRM orientation is 285°, 23°; this direction is close to the one obtained for the coarse grained gabbros, which are believed to be the lithological unit carrying the most stable NRM directions. The corresponding paleopole position is: 202°, 18° (158° W, 18° N).

Curie Temperature

It was thought desirable to study in further detail the properties of the magnetic minerals of representative material from the metabasalts and metagabbros investigated. Thermomagnetic curves were obtained on 5 to 10 mg of powdered rock samples, using a recording electromagnetic microbalance. Most of the curves obtained are relatively simple and indicate the presence of a single magnetic mineral (pure magnetite) having a Curie temperature (Fig. 6) in the 574–582 °C range for the metabasalts (26E11) and the medium grained gabbros (39G11); some mineralized, medium grained gabbros contain magnetic pyrrhotite (θ_c = 310 °C). The fine grained metagabbros have a Curie temperature

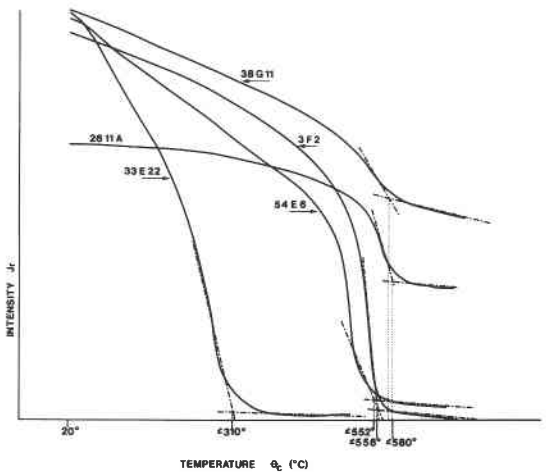


FIG. 6. Typical thermomagnetic curves for the metabasalts and the metagabbros.

range between 558 and 571 °C, suggesting a mixture of 3 to 6% ulvospinel with magnetite (3F2, 5-E6A). In the presence of circulating nitrogen (20 cm^3/min), the pure magnetite and magnetite–ulvospinel mixture depict a perfectly reversible curve, but pyrrhotite does not.

Time (Age) and Origin of Magnetization

Radiogenic age determinations, particularly of Precambrian rocks, are often difficult to interpret and their relationship to paleomagnetic studies have been discussed by a number of authors (Fahrig and Jones 1969; Irving *et al.* 1972). The mean paleopole position of these geological formations does not come close to the one of the Labrador Trough ironstones (149° W, 00°), which are slightly older (Seguin 1972*a*). However, the selective paleopole position (158° W, 18° N) obtained after removal of highly scattered unreliable results is much closer. Fold tests conducted in both natural and demagnetized rock types do not indicate any significant change in the angle of dispersion (Table 2). The residual remanent component is at least partly unrelated to pre-tectonic and/or pre-metamorphic origin of the rocks and partly unrelated to the Hudsonian orogeny. The mean paleopole position obtained by averaging all the cleaned results on the selective pole position after AF demagnetization and correction for the tilt of the formations indicates a late metamorphic event probably related to secondary magnetite as well as pyrrhotite mineralization, assuming that the remanent magnetization (after AF cleaning) was acquired during a single event in the 1200–1300 m.y. bracket (position of the paleopole on the apparent wandering path defined by Irving and Park 1972).

Alternatively, it may be argued, as was done by one of the referees, that the dispersion of the site poles does not represent the direction of a single magnetization acquired at the same time in all the rock formations studied and that the magnetic history of the Labrador Trough rocks is complicated. The remanence acquired at different times at different sites may be in some instances a chemical remanence produced by alteration, and in other instances a partial thermoremanence acquired during burial or a reheating event. Some poles fall on Track 4 of Irving and Lapointe (1975), suggesting an age of 1700 to 1400 m.y., and others lie near the

Logan Loop (Robertson and Fahrig 1971), suggesting an age of about 1000 m.y.

Discussion of the Results

The initial directions of magnetization are quite scattered and this scattering is not substantially reduced in a peak alternating field of 150 Oe. The scattering is attributed to temporary components acquired in the laboratory, viscous components acquired in the Earth's magnetic field, a small fraction of chemical magnetization, secular variation, and measuring errors. A very high proportion of the opaque minerals are small- to medium-sized magnetite crystals, which are considered to dominate the magnetic properties of the rocks. Smaller quantities of pyrrhotite are also present in the fine and medium grained metagabbros. The NRM component is generally less stable and occasionally unstable in pyrrhotite-bearing samples.

Many sites yielded mixed polarities, but the preponderance of normal polarities is significant. Assuming that the remanent magnetization is attributed to a single cause, the mean paleopole position and the correction for tilt suggest that the residual remanent component is not related to the primary thermoremanence at the time of eruption or intrusion or even to the early Hudsonian orogenic event, but rather to the last stage of metamorphism, which appears to be accompanied by mineralization of secondary magnetite and pyrrhotite. The paleopole position would indicate an age of 1250 m.y. for this late geological event.

Magnetic cleaning and restricted thermal treatment did not allow tracing back the original thermoremanent residual component imprinted at the time of intrusion and extrusion of the rocks.

Even after using a method of analysis rejecting the most highly scattered paleomagnetic data, the pole position did not change drastically. It is then possible that the grand average is meaningless, but that the individual results listed in Table 2 may be more useful if their dispersion represents different ages of magnetization.

In spite of the fact that no good paleomagnetic pole can be determined in this work, it contains much valuable information about the petrology, geochemistry, and magnetic characteristics of rocks from a most important part of the Precambrian Shield.

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